Fiscal Year:	FY 2007	**Task Last Updated:**	FY 09/27/2006
PI Name:	Mitchell, Cary Ph.D.		
Project Title:	(NSCORT) for Advanced Life Support at Purdue/Howard/AAMU - Education & Outreach Lead		
Division Name:	Human Research		
Program/Discipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES		
Program/Discipline--Element/Subdiscipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES--Advanced life support		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	cmitchel@purdue.edu	**Fax:**	FY 765-494-0391
PI Organization Type:	UNIVERSITY	**Phone:**	765-494-1347
Organization Name:	Purdue University		
PI Address 1:	Department of Horticulture and Landscape Architecture		
PI Address 2:	625 Agriculture Mall Dr		
PI Web Page:	http://www.alsnscort.org		
City:	West Lafayette	**State:**	IN
Zip Code:	47907-2010	**Congressional District:**	4
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	NSCORT
Start Date:	12/01/2002	**End Date:**	11/30/2006
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: End date is 11/30/2006, per info from PI office (2/2009)		
Key Personnel Changes/Previous PI:	Previous E&O Co-I changed positions outside the ALS NSCORT.		
COI Name (Institution):			
Grant/Contract No.:			
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The ALS NSCORT did not have a formal E&O program during this abbreviated last official year.		
Rationale for HRP Directed Research:			

Research Impact/Earth Benefits:	In past years the Education and Outreach programs goals were to: • Increase the number of students/educators who are involved in NASA related education opportunities. • Increase the scientific literacy of K-12 educators, students and general public, fostering public awareness and the earth benefits of NASA research. • Increase the number of students directly engaged in NASA research. • Increase the number and diversity of students who pursue graduate degrees and research careers in NASA related science, engineering and technology fields. • Increase higher education research capability and opportunities that attract and prepare students for NASA related careers.
Task Progress:	E&O experienced very limited activity due to key staff turnover. The ALS NSCORT's Director and Operations Manager continued to respond to requests from both adult and student organizations. Invited Presentations and Requests 2006 • 2/28/06: Dave Kotterman presented “Exploring: How Will We Live on Other Planets?” at the Blue and Gold Boy Scout Banquet at Burnett Creek Elementary School. 75 total scouts and their parents were in attendance. • 3/14/06: Dave Kotterman presented “Exploring: How Will We Live on Other Planets?” at Burnett Creek Elementary School for 25 fifth graders and their teacher. • 4/26/06: Cary Mitchell and Dave Kotterman presented to the Wilmette Harbor Rotary Club, Wilmette, Illinois at the Sheridan Shore Yacht Club. The presentation entitled “Artificial Closed Eco-Systems for Extended Human Habitation in Space” was well received by 40 people. • 5/9 – 5/10/06: Cary Mitchell served as a Special Awards Judge for the Intel – sponsored Science and Engineering Fair in Indianapolis. This annual event acknowledges and motivates science and engineering career aspirations in middle and high school students. • 8/23/06: Cary Mitchell presented a talk entitled “Artificial Closed Ecosystems for Long-Duration Human Habitation of Space” to the Westminster Village Retirement Community. 50 people in attendance. • 8/24/06: Cary Mitchell and Gioia Massa presented a seminar for the Department of Horticulture and Landscape Architecture entitled “Artificial Closed Ecosystems for Long-Duration Human Habitation of Space”. 35 people in attendance. SUPPLEMENTAL INFORMATION (provided in 2007): Presentations given during 2007 are: • Cary Mitchell and Gioia Massa attended the Indiana Horticultural Congress on January 31 in Indianapolis where Cary Mitchell presented “Future technologies for high tunnels” co-authored by Gioia Massa. • Cary Mitchell made a presentation entitled “Sustaining Human Life in Space with Help from Plants and Bioprocessing” at the Mid-America Collegiate Horticulture Society conference held at Purdue on March 9th. • Cary Mitchell and Gioia Massa attended the NCERA-101 committee on Controlled Environment Technology and Use from April 14-17 at Tuskegee University, where Cary Mitchell presented the Purdue Station Report authored by Gioia Massa, Rob Eddy, and Cary Mitchell. • Cary Mitchell gave a presentation entitled “Artificial Closed-Loop Ecosystems for Human Habitation of Space” to the Purdue University “Students for the Exploration and Development of Space” on April 12 • Gioia Massa presented a lecture on ‘Plants in Space” to a Purdue University Summer Program on July 3rd. • Cary Mitchell, Gioia Massa, and Craig Schluttenhofer attended the American Society for Horticultural Science annual conference in Scottsdale Arizona July 16-19th and Gioia Massa participated as a workshop speaker in a workshop on “Light emitting Diodes for Horticulture” presenting “Plant Responses to LED Lighting” co-authored by Gioia Massa, Hyeon-Hye Kim, Ray Wheeler, and Cary Mitchell. • Cary Mitchell gave presentations entitled “How Controlled Environments will Enable Regenerative Agricultural Systems in the Twenty-First Century” and “Mechanical Stress Regulation of Plant Growth and Development” to the Department of Environmental Sciences at the University of Toledo, OH on August 31st. • Cary Mitchell and Gioia Massa attended the American Society for Gravitational and Space Biology *ASGSB) annual meeting October 25-28th, 2007 at Ames Research Center where Gioia Massa presented “Testing of the HELIAC 2 Plant Detection and Lighting System.” This presentation was co-authored by C. Michael Bourget and Robert Morrow of ORBITEC and Cary Mitchell.
Bibliography Type:	Description: (Last Updated: 04/16/2020)